
Statistics 212: Lecture 24 (April 30, 2025)

Diffusion Sampling

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1 Recap of Last Class

Theorem 1.1 (Cameron–Martin). *Let μ be a deterministic function $\mu : [0, T] \rightarrow \mathbb{R}$ such that $\int_0^T \mu(t)^2 dt < \infty$. Denote by $\mathbb{P}$ the Wiener measure on $C([0, T]; \mathbb{R})$ induced by a standard Brownian motion $(B_t)_{t \in [0, T]}$, and let $\mathbb{P}^\mu$ be the law of the process $B_t^\mu = B_t + \int_0^t \mu(s) ds$. Then $\mathbb{P}^\mu$ is the law of $B_t + \int_0^t \mu(s) ds$ and satisfies $\frac{d\mathbb{P}^\mu}{d\mathbb{P}}(B_{[0, T]}) = \exp(\int_0^T \mu_s dB_s - \int_0^T \mu(s)^2 / 2 ds)$. This is a local martingale.*

Corollary 1.2. *If $f(t) = \int_0^t \mu_s ds$ has an L^2 derivative then it follows that $\text{Law}(B_t)$ and $\text{Law}(B_t + f_t)$ are mutually absolutely continuous.*

The next result is a converse. (Often the “Cameron-Martin theorem” means the conjunction of the corollary above and the theorem below.)

Theorem 1.3. *Conversely, if $f' \notin L^2$ then $\text{Law}(B_t + f_t)$ is singular with respect to $\text{Law}(B_t)$, meaning that there exists a set $A \subset C([0, T])$ such that $\mathbb{P}_\mu(A) = 1$ and $\mathbb{P}_0(A) = 0$.*

Proof. (Sketch) The idea is to project on to a finite dimensional Fourier series subspace.

WLOG, set $T = 1$, $f(0) = 0$, and $f(1) = 0$. Now consider the Fourier coefficients of a Brownian Bridge $\tilde{B}_t = (B_t - tB_1)$.

$$\tilde{B}_t = \sum_{k=1}^{\infty} c_k \sin(k\pi t), \quad c_k \sim N\left(0, \frac{a}{(k\pi)^2}\right) \text{ independent.}$$

for some appropriate constant a . Similarly, Likewise expand $f(t) = \sum_{k=1}^{\infty} d_k \sin(k\pi t)$, so that $f'(t) = \sum_{k=1}^{\infty} k\pi d_k \cos(k\pi t)$ and $\|f'\|_{L^2}^2 = \int_0^1 f'(t)^2 dt \propto \sum k^2 d_k^2$. Now note that if f has Fourier coefficients $d_1, d_2, d_3, \dots$ while f' has Fourier coefficients $d_1, 2d_2, 3d_3, \dots$. On the finite-dimensional projection onto modes $1, \dots, n$, notice that $\text{Law}(c_1, \dots, c_n) \propto \exp(-\alpha \sum_{k=1}^n k^2 c_k^2)$. This sum is a (non-standard) inner product of $\langle \vec{c}, \vec{c} \rangle$. Then, notice that $\sum k^2 d_k^2 = 0 \iff \|\vec{d}\|^2 = \langle \vec{d}, \vec{d} \rangle = 0$. Shifting by $(d_1, \dots, d_n)$ yields absolute continuity if and only if $\sum_{k=1}^{\infty} (k\pi)^2 d_k^2 < \infty$, i.e. $f' \in L^2$.

If the sum diverges, the measures become singular in the limit. In particular, on the finite-dimensional projection onto modes $1, \dots, n$, the law of

$$(c_1, \dots, c_n) \sim N\left(0, \text{diag}((k\pi)^{-2})_{k=1}^n\right)$$

has density

$$\propto \exp\left(-\frac{1}{2} \sum_{k=1}^n (k\pi)^2 c_k^2\right).$$

Shifting by the vector $(d_1, \dots, d_n)$ yields two Gaussian measures which are equivalent if and only if $\sum_{k=1}^{\infty} (k\pi)^2 d_k^2 < \infty$, i.e. $f' \in L^2([0, 1])$. If $\sum k^2 d_k^2 = \infty$, then in the limit $n \rightarrow \infty$ you obtain a sequence of sets on which one measure puts mass $\rightarrow 1$ and the other $\rightarrow 0$, proving mutual singularity. $\square$

2 Diffusion!

2.1 Sequential Sampling and Polya's Urn

Suppose my goal is to sample from some probability measure on $\mu \sim P([0, 1])$. We define the law μ_{seq} on infinite sequences $(X_1, X_2, \dots)$ by first taking $p \sim \mu$, and then we have $X_1, X_2, \dots \stackrel{iid}{\sim} \text{Bern}(p)$, which are in fact correlated since p is unknown.

Claim: If I can sample from this sequence measure I can also sample from μ , since $\lim_{T \rightarrow \infty} \frac{X_1 + \dots + X_T}{T} = p$ almost surely. Now we can sample μ_{seq} without first choosing p . Steps:

- Sample $X_1 \sim \text{Law}_{\mu_{\text{seq}}}(X_1) \sim \text{Bern}(\mathbb{E}^{\mu}[p])$
- Sample $X_2 \sim \text{Law}_{\mu_{\text{seq}}}(X_2 | \mathcal{F}_1) \sim \text{Bern}(\mathbb{E}^{\mu}[p | \mathcal{F}_1])$
- ...
- Sample $X_k \sim \text{Law}_{\mu_{\text{seq}}}(X_k | \mathcal{F}_{k-1})$
- ...

Where we define $\mathcal{F}_{k-1} = \sigma(X_1, \dots, X_{k-1})$. We have μ_{all} for the law of $(p, X_1, X_2, \dots)$ and we have $\text{Law}_{\mu_{\text{all}}}(X_1, X_2, \dots | p) \stackrel{iid}{\sim} \text{Bern}(p)$. Then $\text{Law}_{\mu_{\text{all}}}(X_1, \dots) = m_{\mu_{\text{seq}}} = \int \text{Bern}(p)^{\otimes \infty} d\mu(p)$

Slogan: To sample I just need to be able to compute conditional means. The idea is to do this for a LOT of our X_i s in this way and then we have that $p \approx \frac{X_1 + X_2 + \dots + X_T}{T}$

2.1.1 Polya's Urn

Consider an urn that starts with 1 black ball and 1 white ball. At each step, choose 1 ball uniformly, replace it with a copy. Suppose we have b black balls and w white balls in the urn then $P(\text{next ball is black}) = \frac{b+1}{b+w+2}$ in the next step. This is identical to sequential sampling as above with $\mu \sim \text{Unif}(0, 1)$. In particular, let $X_i = 1$ if the i th ball is black and 0 if it is white. We begin by sampling $X_1 \sim \text{Bern}(\mathbb{E}^{\mu}[p]) = \text{Bern}(\frac{1}{2})$. Suppose we pick a black ball, next. Then $X_1 = 1$, and so $\mathbb{E}(p | X_1) = 2/3$. Then, we sample $X_2 \sim \text{Bern}(\mathbb{E}^{\mu}[p | \mathcal{F}_1]) = \text{Bern}(2/3)$. We can continue in the way described above. If k of the $X_1, \dots, X_n$ are black (ie $= 1$). Then we have that $\text{Law}(p | \mathcal{F}_n) \propto p^k (1-p)^{n-k}$, which means it has a Beta distribution. By beta properties we have $\mathbb{E}(p | \mathcal{F}_n) = \frac{k+1}{n+2}$.

This whole equivalence story means that the sequence $(b, w, w, b, b, w, \dots) \sim \mu_{\text{seq}}$ for $\mu \sim \text{Unif}(0, 1)$. The limiting fraction of b is $\text{Unif}(0, 1)$.

2.2 Diffusion Sampling from the Perspective of Stochastic Localization

We'd now like to apply this discretized approach to continuous distributions.

Goal Sample $X \sim \mu \in P(\mathbb{R}^d)$ where μ is compactly supported etc. This might be computationally hard when d is big (unlike the Polya urn example). Instead of Bernoulli observations, use Gaussian noisy observations and we'll take a small step limit.

Consider a small ϵ signal to noise ratio per step. Given the unknown X , observe

$$w_1 \sim \epsilon X + Z_1$$

$$w_2 \sim \epsilon X + Z_2$$

where $Z_i \stackrel{iid}{\sim} N(0, 1)$. We apply the same sequential sampling procedure. Here $\mu_{all} = \text{Law}(x, w_1, w_2, \dots)$ while $\mu_{seq} = \text{Law}(w_1, w_2, \dots)$. Now we can sample μ_{seq} without first choosing X . We generate one sample as follows:

- Sample $W_1 \sim \epsilon \mu * N(0, \epsilon)$
- Sample $W_2 \sim \epsilon \text{Law}(x | \mathcal{F}_1) * N(0, \epsilon)$
- ...
- Sample $W_k \sim \text{Law}(X | \mathcal{F}_k) * N(0, \epsilon I_d)$
- ...

As before $X \approx \frac{W_1 + W_2 + \dots + W_T}{T\epsilon}$

As written, this is tricky to compute because we have to sample each W_k from a convolution. But you can pass to the continuous time $\epsilon \rightarrow 0$ limit and things get nicer. Here for the “known x ” process we keep track of $y_{k\epsilon} \approx \sum_{j=1}^k W_j \sim k\epsilon x + N(0, k\epsilon)$. Hence $W_k | X \sim \mathcal{N}(\epsilon X, \epsilon I_d)$.

After T steps one recovers $\hat{X}_T = \frac{1}{T\epsilon} \sum_{k=1}^T W_k \rightarrow X$ a.s.

In the limit, $\epsilon \rightarrow 0$, first notice that conditionally on X , you get $y_t = w_t | X \sim BM$ with drift X and $dy_t = dB_t + X dt$.

...

First, recall the fact that $\mu_{path} = \text{Law}(Z_t)$

$$dZ_t = dB_t + m_t dt$$

and from the Cameron Martin Theorem, we have $m_t = \mathbb{E}^{\mu_{all}}[X | Y_t = Z_t] = \frac{\int x e^{\langle X, Z_t \rangle (-tx^2/2)} d\mu(x)}{\int e^{\langle X, Z_t \rangle (-tx^2/2)} d\mu(x)}$

Q: Why are Y_t and Z_t equal in distribution?

Consider the process y_t relative to its natural filtration $\mathcal{G}_t$, where $(X \text{ unknown})$. Then we have that $y_t - \int_0^t \mathbb{E}[X | \mathcal{G}_s] ds$ is a $\mathcal{G}_t$ martingale. In fact M_t is Brownian Motion with respect to $\mathcal{G}_t$. To see why, note first that Dubins-Schwartz tells us M_t is a time change of Brownian motion. We know that the time change must be trivial, since if there were non-trivial time change, then the quadratic variation “clock” $\lim_{\delta \downarrow 0} \sum (X_{(k+1)\delta} - X_{k\delta})^2$ would be wrong.

2.3 Standard Diffusion

One can write down the corresponding partial differential equation for the evolving density, or compare to the usual “score-based” diffusion samplers by re-parameterizing time and variance (see references).

2.4 References

For more on diffusions along the lines of the above approach, see [Sampling, Diffusions, and Stochastic Localization](#) by Andrea Montinari (<https://arxiv.org/abs/2305.10690>). (Many many **many** other good diffusion references as well.)